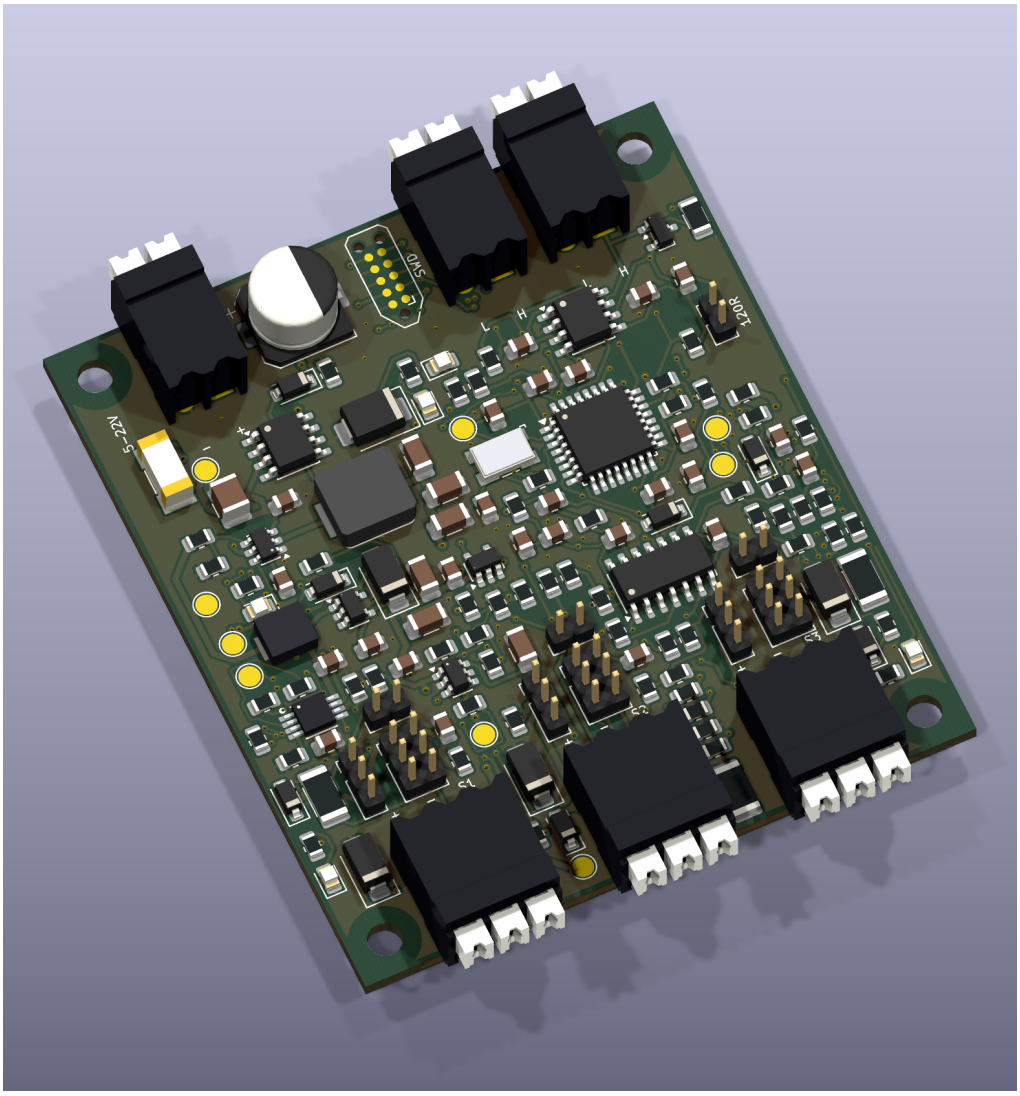
FRC
Sensor CAN

Revision B-2 · Prototype engineering and assembly guide

Three industrial / logic sensor inputs
Protected battery and regulated 5 V sensor power



2026-09-08

Editable KiCad files, turnkey assembly BOM, firmware, robot examples and verification records accompany this guide. No physical prototype has been tested. Electrical ratings are design targets pending the included qualification plan.

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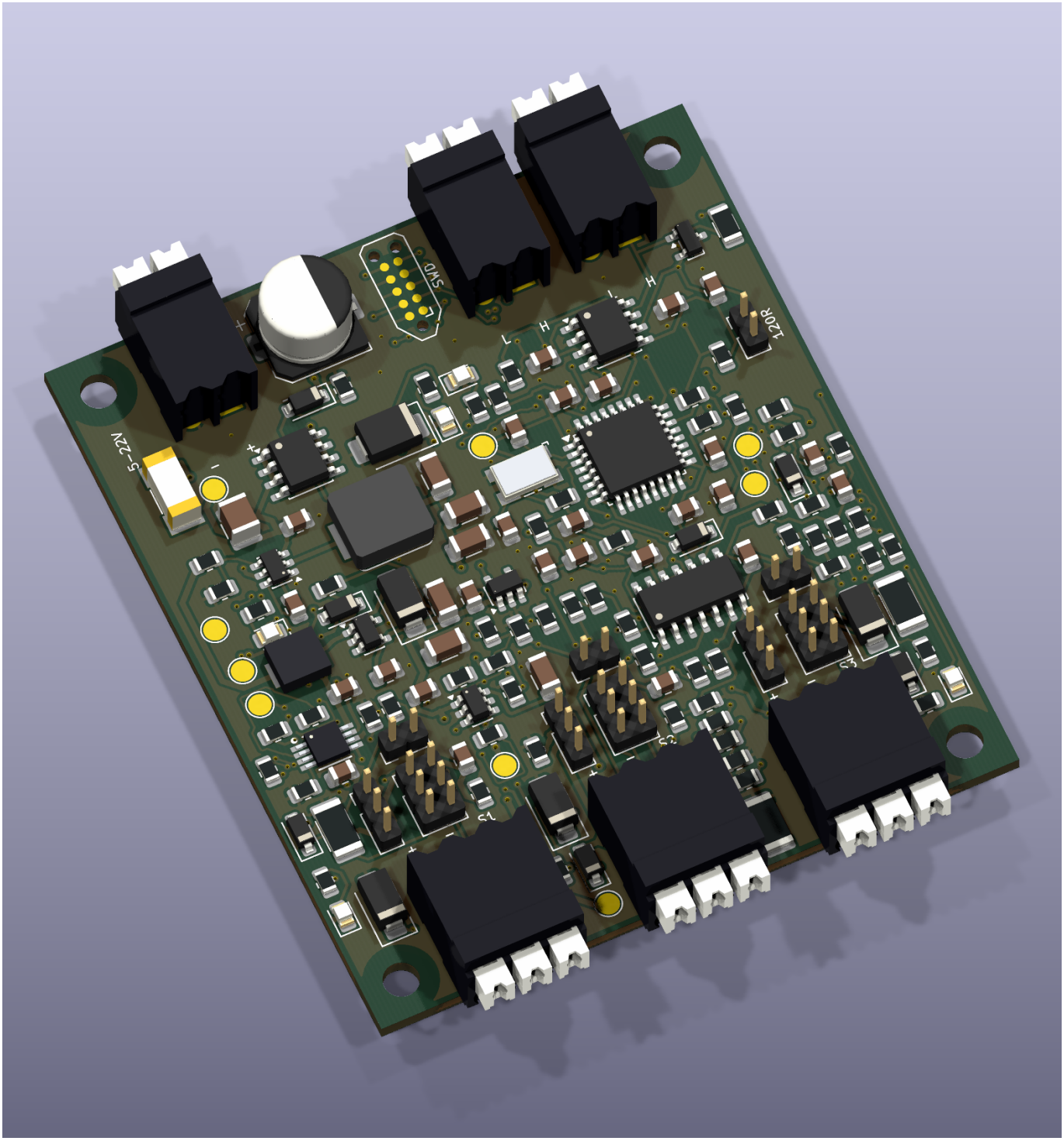
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1 Product and delivery status

Source: README.md

Scope: Rev B-2. Revision overview · All revisions.

China factory-assembly variant of the horizontal-connector Rev B. Three sensor inputs, selectable regulated 5 V or battery sensor power, a 5–22 V board-input target and a 60 × 68 mm PCB. All 144 components mount on top. This is an unbuilt prototype with a complete quotation/build-data package; factory sourcing, panel engineering and placement review are still required.



The board uses horizontal **Weidmüller 1824420000** (power/CAN, two poles) and **1824430000** (sensors, three poles), LSF-SMT 3.50 **/90**. Reserve 60 × 76.1 mm for the assembly before cable space; the terminals are 8.5 mm high.

B-2 changes 37 resistors and 12 bypass capacitors to checked equivalents, and buys eight unchanged MPNs from cheaper observed sources. Nominal consumed component cost falls from $59.30 to **$46.35**, before MOQ, factory fees and shipping. No parts were deleted. The circuit, copper geometry, connector pinouts and firmware remain equivalent to Rev B.

Order and cost

- Factory upload packet
- Factory order instructions, Gerber ZIP
- Complete turnkey BOM, JLCPCB matching BOM, factory CPL, loose accessories
- Estimated order costs, cost spreadsheet, MOQ purchase quantities
- Exact part changes, capacitor audit

Budget roughly **$300–390 total for two assembled boards**, or **$435–535 for five**, including estimated shipping and excluding import taxes and optional factory programming. JLCPCB publishes a two-board minimum. These are engineering estimates, not a live quotation; the MCU and connectors require procurement review. This prototype is not yet a competitive retail-cost design.

Engineering files

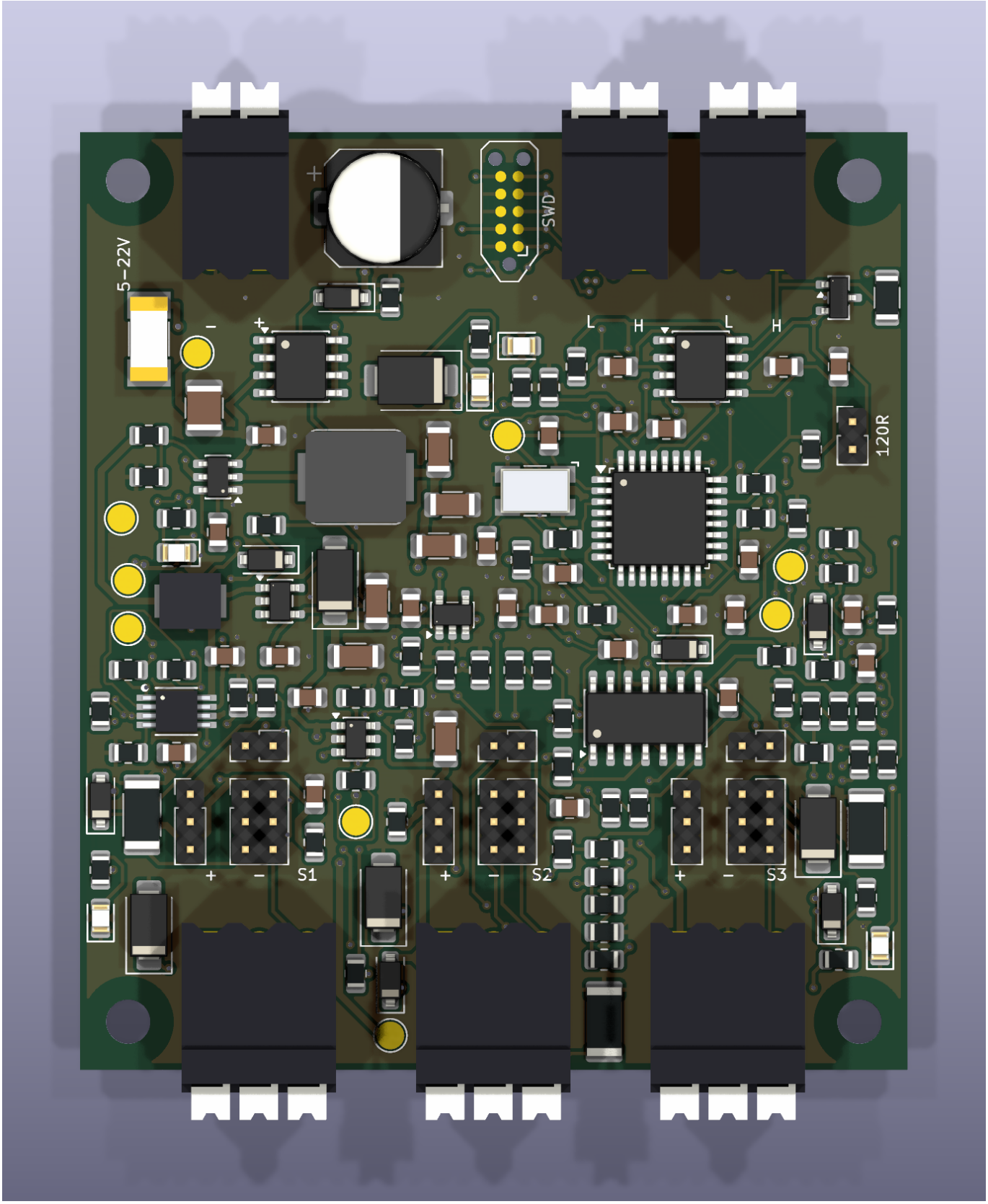
- KiCad project, schematic, PCB
- Assembly drawing, schematic PDF, interactive BOM, STEP
- Printable handbook, printable BOM
- Wiring and configuration, firmware/programming, robot integration, validation, qualification worksheet

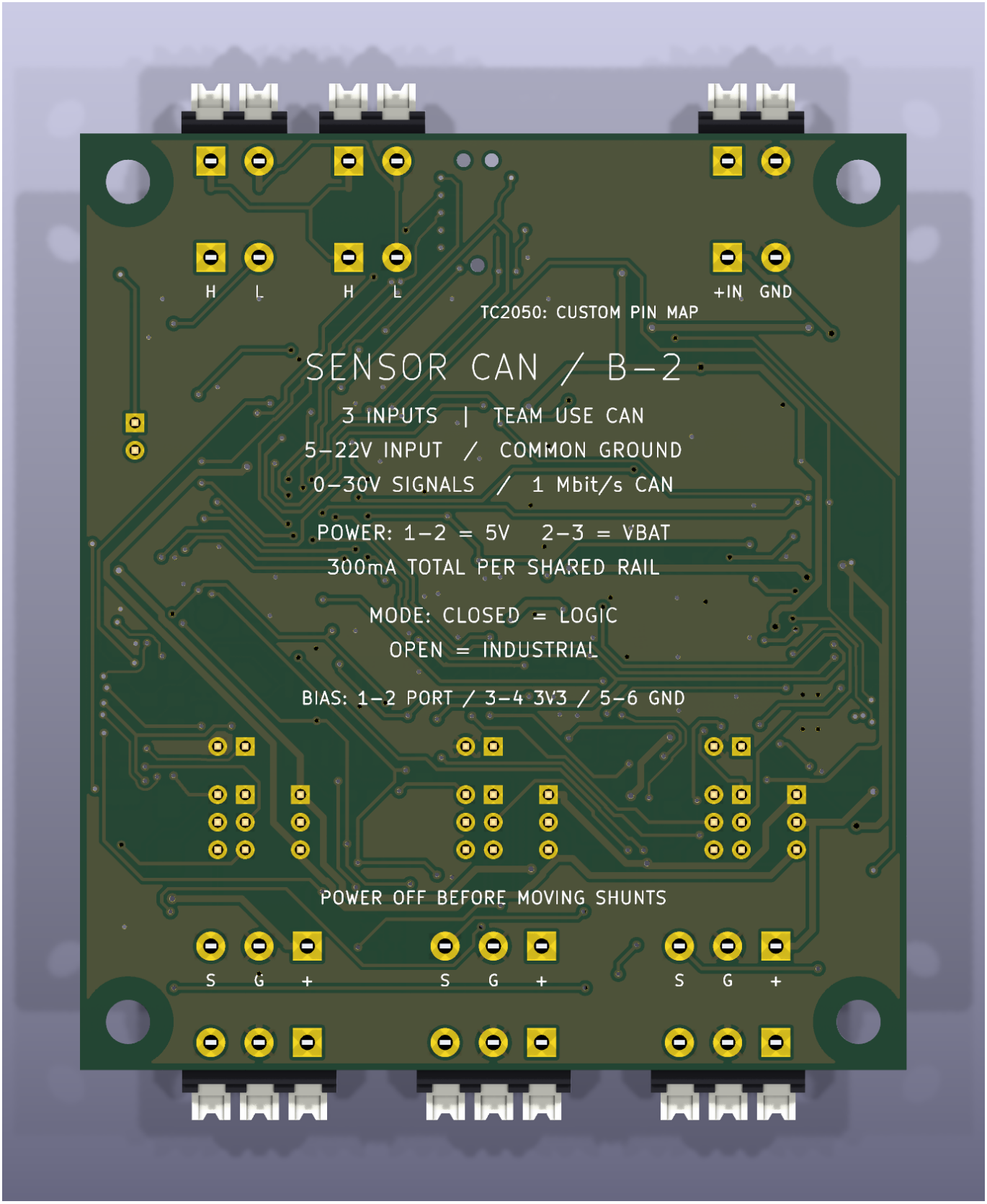
The input range is a design target measured at the board connector under load. VBAT follows the input and is not regulated 12 V. Shared sensor budgets are 300 mA on 5 V and 300 mA on VBAT, not per-port. A low battery can stop an industrial sensor with a 10 V minimum supply. Signal inputs accept positive 0–30 V through the protected divider with common ground; logic-mode targets are LOW ≤0.8 V / HIGH ≥2.4 V. The design does not promise 1.8 V logic or CTRE Phoenix/Tuner compatibility.

Native and exported-data checks are computational evidence. Actual thermal, surge, CAN, sensor and programmer tests remain pending. All shunts ship loose and all sensor-power selectors open. No physical board, purchase or factory upload has been made.

See tools, sources and factory requirements before manufacturing.

Additional board views





Connector housings use official manufacturer STEP geometry; see connector verification. Render colors are illustrative.

2 Factory order instructions

Source: docs/FACTORY_ORDER.md

Scope: Rev B-2. Revision overview · All revisions.

This package is ready to submit for a **prototype fabrication and turnkey assembly quotation**. Factory stock matching, panel/fixture review and placement-orientation approval remain necessary before manufacture. It is not a completed JLCPCB checkout or a qualified production release.

Files to submit

1. `order/RevB2_Gerbers.zip`: single-board Gerbers and plated/nonplated Excellon drills, in millimeters. Do not scale or mirror.

2. `bom/BOM_turnkey.csv`: complete exact-manufacturer/MPN sourcing BOM, including all 144 fitted parts and ten loose shunts per board.

3. `output/assembly/CPL_JLCPCB.csv`: all 144 fitted placements, top side. `bom/BOM_JLCPCB.csv` supplies the corresponding 50 fitted BOM groups. Blank LCSC fields require exact-MPN global sourcing or consignment; they do not mean DNP. The shunts are separately listed in `bom/ACCESSORIES.csv` and must be bagged.

4. `output/assembly/top_assembly.pdf`, `bottom_markings.pdf`, `schematic.pdf`, `pad_coordinates.csv` and the interactive BOM for pin-one and polarity review.

5. `output/mechanical/FRC_Sensor_CAN.step`, the 1:1 mechanical PDF, and these instructions. The STEP models are conservative placement aids, not certified connector tooling models.

PCB and panel requirements

Setting	Requirement
Finished board	60 × 68 mm, 2 layers, 1.6 mm FR-4
Copper	Nominal finished 35 µm / 1 oz per side; use the supplied electrical clearances
Finish	ENIG, both masks, green mask, white silkscreen
Minimum features	0.18 mm tracks; 0.20 mm clearance; ordinary vias 0.60/0.30 mm; preserve every supplied drill
Assembly	128 SMT and 16 through-hole components, all on top; no bottom components
Stencil	0.10 mm starting specification, supplied top paste; assembler to review aperture release and U3 exposed-pad paste
Pogo connector J8	Ten bare ENIG contacts: NO paste, solder, connector or conformal coating; three alignment holes remain NPTH
Mounts	Four 3.2 mm NPTH; do not fill or plate
Solder	Lead-free process, engineer-reviewed profile compatible with every supplied component

For JLCPCB **Standard PCBA**, the single PCB is below the published 70 × 70 mm minimum processing envelope. Request factory panelization with removable tooling rails and fiducials. A 5 mm rail allowance around each side gives a nominal **70 × 78 mm processing outline** before panel spacing/tabs. The assembler must finalize tooling-hole/fiducial/tab positions. Do not drill new tooling holes in the functional 60 × 68 mm board: there is routed copper near the edges.

The six horizontal connector bodies project beyond the top and bottom board edges. Assemble SMT while supported by rails, then depanel and install these connectors in a supporting fixture, or propose an equivalent process with sufficient overhang clearance. The finished assembly reserves **60 × 76.1 mm**, excluding cables; terminal height is **8.5 mm**, leads **3.5 mm** before trimming. Breakaway tabs and fixtures must not foul the

housings, actuator travel, solder joints or nearby copper. Panel design is part of the factory's engineering service, not a supplied production panel.

The selected Samsung 100 nF capacitor's reference sheet recommends a maximum 250 °C reflow peak for six seconds. Use a controlled profile within component limits (JLCPCB Standard lists 240 ±5 °C). Do not automatically apply Economic PCBA's listed 255 ±5 °C profile to this BOM. Solder THT parts afterward. U3's exposed thermal pad must be soldered and inspected; do not omit it or its thermal vias. The assembler must review paste loss into these vias; no via-in-pad filling process is presumed in the cost estimate.

Placement coordinates and orientation

The factory CPL origin is the **lower-left finished-board corner viewed from above**: X increases right, Y increases up, and all coordinates are positive millimeters. Rotations are KiCad top-side counterclockwise degrees. SMT parts use the native body-center origin. Through-hole header and terminal centers have explicit offsets from their KiCad pin-one origins; those offsets are recorded in `coordinate_offsets.csv`.

`pad_coordinates.csv` provides every numbered pad in the same coordinate frame and is the final reference for matching factory library orientation. Confirm U1–U8, Q1, Y1, every diode/LED and C1 polarity. Do not assume an LCSC library's zero-degree rotation matches KiCad. The original `positions_all_top.csv` uses KiCad's upper-left origin with negative Y and is provided only as a native reference; **do not mix it with the factory CPL**.

Parts that require particular attention

• J1–J3: **Weidmüller 1824420000**, LSF-SMT 3.50/02/90 3.5SN BK TU. J4–J6: **1824430000**, LSF-SMT 3.50/03/90 3.5SN BK TU. These are horizontal wire entry, parallel to the PCB. Two legs per pole, 7 mm between rows, 1.1 mm drills. Do not substitute `/135`, `/180`, Phoenix or an unreviewed clone.

• U5: **STM32F042K6T6**, LQFP32. LCSC C69216 was out of stock at the observation date. Procure from DigiKey or another authorized source. Do not substitute STM32F030 or another MCU; pin compatibility alone is insufficient.

• U7: **LM339LVDR**, not an ordinary LM339. Its low-voltage input behavior is required.

• U4: **TPS2553DBVR-1**, including the `-1` latch-off suffix. U3: **TPS1H200AQDGNRQ1** with its exposed pad.

• Keep the specified inductors, fuse, TVS diodes, clamp diodes, timing/filter capacitors and crystal exact. Only the five part-family substitutions listed in `PART_CHANGES.md` are incorporated. Quote any further proposed change for engineering review before fitting it.

Every BOM reference is fitted unless explicitly labeled an accessory. J8, test pads and mounting holes are unpopulated PCB features. Deliver all ten configuration shunts loose in an antistatic bag. Ship power selectors, bias selectors, threshold selectors and CAN termination **open**. This leaves sensor-power connections open until the user configures the board.

Inspection, optional programming and delivery

Include bare-board electrical testing, SMT AOI, THT visual inspection, polarity inspection, and inspection of U3's hidden solder joint. Check that paired terminal legs connect to the same pole and adjacent poles remain isolated. Do not count bare-board testing/AOI as a functional CAN or sensor test.

The base assembly quote may ship an unprogrammed MCU. For a programmed board, separately quote flash/verify and basic power checks using `firmware/build/sensor_can.hex` or `.bin` (address **0x08000000** for BIN). File hashes are in the archive manifest. Read `firmware/README.md` and `WIRING.md`: J8 is a TC2050-IDC-NL **mechanical** pattern with a **custom electrical pinout**. Its pin 6 is target-voltage reference, not programmer power. Power the board through J1 with a current-limited supply; never inject probe 3.3 V into J8.

With all shunts removed, check the intermediate rail (~3.97 V), internal 5 V and logic 3.3 V. Flash, read back, verify and leave sensor rails disabled. Leave flash configuration erased unless an explicit per-unit ID provisioning list is supplied. A production fixture, full CAN/sensor/load tests and qualification are additional work; `QUALIFICATION.md` describes the required engineering measurements.

Pack assemblies individually in ESD bags, protect actuator overhang and pogo pads, include loose shunts and unused boards/components as agreed in the quotation. Return the exact fitted-MPN list and programming/inspection results with each batch.

Sources: JLCPCB assembly capabilities and reflow temperatures, JLCPCB pricing, Samsung capacitor reference sheet.

3 Cost and procurement

Source: docs/COST.md

Scope: Rev B-2. Revision overview · All revisions.

These are USD engineering estimates dated 2026-09-08, not a submitted quotation. The PCB has not been physically qualified. Pricing assumes full SMT and through-hole assembly, ENIG, bagged shunts and ordinary international shipping; destination taxes and tariffs are excluded.

JLCPCB publishes a minimum of **two assembled boards**. If you only need one, the first realistic JLCPCB order still pays for two. The one-board row in the CSV is hypothetical and must not be treated as an available service. PCBWay's current capability page publishes a five-piece minimum.

Assembled boards	Estimated order total	Approx. per board
2	$300–390	$151–194
5	$435–535	$87–107
10	$665–800	$67–80
25	$1380–1630	$55–65

The component-only consumed comparison is **$46.35 per board versus Rev B's $59.30**, about 22% lower on the stated first-tier basis. This comparison does not include MOQ leftovers or assembly. It combines cheaper equivalents with cheaper procurement of identical parts; it is not a $46 finished-board price.

The order model uses dated per-line prices and MOQ rounding, then adds the published JLCPCB Standard setup/stencil/feeder and solder-joint rates. It conservatively budgets feeder fees for all 50 fitted part types. Other costs are explicit allowances: ENIG boards/panel rails, attrition, procurement handling/inbound freight, inspection and international shipping. The factory's actual sourcing quote may differ considerably. LCSC inventory is not JLCPCB assembly inventory.

Factory programming and simple no-load checks are optional and additional: budget **$50–100 setup plus $3–5 per board**, including an allowance for the custom SWD adapter. The archive supplies the firmware and wiring. Full sensor/CAN/load testing, a repeatable production fixture and qualification are separate work.

At prototype quantities, this is not yet a cost-competitive retail product. The 50 fitted part types, premium connectors and supplier setup costs dominate; reducing capacitors alone does little. A further cost revision should consolidate functional circuits and selectors after this prototype is measured.

See `bom/cost_estimate.csv` for every allowance and `bom/procurement_quantities.csv` for exact modeled purchase quantities, leftover parts and price breaks.

Sources: JLCPCB assembly fees, JLCPCB prototype minimum, PCBWay minimum. Component source URLs and observation dates are in the BOM.

4 Component changes

Source: docs/PART_CHANGES.md

Scope: Rev B-2. Revision overview · All revisions.

Rev B-2 changes procurement while preserving Rev B's circuit, 60 × 68 mm outline, horizontal connectors, copper routing, pin assignments and firmware. There are still 144 fitted components and ten loose shunts. No protection, filtering or decoupling part has been deleted.

Rev B part	Rev B-2 fitted part	Count	Review basis
Yageo RC0805FR-071KL	UNI-ROYAL 0805W8F1001T5E, C17513	7	1 kΩ, 0805, 1%, 125 mW, ±100 ppm/°C
Yageo RC0805FR-0710KL	UNI-ROYAL 0805W8F1002T5E, C17414	16	10 kΩ, same package/rating/tolerance/TCR
Yageo RC0805FR-074K7L	UNI-ROYAL 0805W8F4701T5E, C17673	7	4.7 kΩ, same package/rating/tolerance/TCR
Yageo RC0805FR-07100KL	UNI-ROYAL 0805W8F1003T5E, C149504	7	100 kΩ, same package/rating/tolerance/TCR
Yageo CC0805KRX7R9BB104	Samsung CL21B104KCFNNNE, C28233	12	100 nF, X7R, 10%, 0805; voltage rating increases from 50 V to 100 V

The resistor family is rated 150 V working voltage, with power derating above 70 °C. Original per-reference power and 1.75% total resistance budgets remain applicable to these 1%/100 ppm parts. The Samsung capacitor is 2.00 × 1.25 × 1.25 mm nominal and fits the existing 0805 lands; maximum height remains below the terminal envelope. Its effective capacitance, reflow profile and real circuit transients still need prototype validation. Changing rated voltage alone never establishes a converter's stability.

The original Yageo MPNs remain approved fallback parts if the assembler cannot source these alternatives. A fallback changes cost and must be recorded in the fitted BOM. No resistor value or dielectric may be changed as an informal substitution.

Eight additional MPNs stay electrically and physically identical but use cheaper observed LCSC procurement: TCAN3413DR, LMR51420XDDCR, TPS613222ADBVR, TPS2553DBVR-1, BAT46W-E3-18, CL31A226KAHNNNE, AO4485 and Littelfuse SMAJ30A. Remaining parts retain their exact DigiKey sourcing references. LCSC stock observations are not reservations or JLCPCB assembly-stock confirmations.

Keeping 0805 parts avoids a needless reroute and remains machine-assembly compatible. 0402 parts would not reduce the number of placements or distinct feeder setups. The premium Weidmüller connectors are retained as requested. Their DigiKey first-tier total is $11.25 per board; they are a substantial part of the cost.

There are 34 capacitors. Most are local bypass, regulator reservoir, crystal load, input filter or fault/reset timing components. C13, C24 and C42 are possible **bench evaluation** candidates for later reduction, not approved DNPs. See `PASSIVE_REVIEW.md` for every capacitor's purpose. Larger savings require functional circuit consolidation and hardware testing.

Primary specification sources: UNI-ROYAL chip resistor family, Samsung exact capacitor reference sheet. Supplier IDs, source URLs, observed quantities, MOQ and prices are recorded in `bom/catalog_china.json` and the turnkey BOM.

5 Wiring and selectors

Source: docs/WIRING.md

Scope: Rev B-2. Revision overview · All revisions.

This revision has three independent digital sensor inputs, two shared switched power rails, and a custom 1 Mbit/s CAN interface. It is a prototype design; physical bring-up and qualification remain outstanding. It does not implement CTRE Phoenix or Tuner.

Connector	Pin 1	Pin 2	Pin 3
J1 input	+5–22 V DC	Ground	—
J2, J3 CAN	CAN H	CAN L	—
J4 sensor 1	Selected power	Ground	Signal
J5 sensor 2	Selected power	Ground	Signal
J6 sensor 3	Selected power	Ground	Signal

Use the numbered pads and board markings. J1–J3 face the opposite board edge from J4–J6, so pin numbers are more reliable than assuming the same left-to-right order. The two solder legs for each terminal pole are the same electrical pin. CAN terminals do not provide a ground pin: power grounds establish the shared reference.

Supply 5–22 V **at J1 under load**. The maximum is 22 V including supply tolerance. A nominal 24 V industrial supply is outside this input rating, although a separately powered sensor may present a positive signal up to 30 V. Use an appropriate fused FRC distribution output and common ground; the onboard 1.5 A fuse does not replace robot distribution protection.

Weidmuller /90 terminals take wires horizontally, parallel to the PCB, with push-in actuators. Sensor terminals are 1824430000; power/CAN terminals are 1824420000. Strip 8 mm. Follow the manufacturer's conductor/ferrule limits; 20–22 AWG flexible robot wiring is a practical starting point. Press the actuator to insert or remove fine-stranded wire when required. Secure cables against robot vibration and pulling. Assemble the through-hole terminals after the surface-mount parts.

Select sensor power

Each port gets its own three-pin selector. **Turn board power off before changing shunts. Fit exactly one power shunt per powered port.**

Port	Selector	Bridge pins 1–2	Bridge pins 2–3	No shunt
1	JP101	Regulated 5 V	Protected switched VBAT	Power pin disconnected
2	JP201	Regulated 5 V	Protected switched VBAT	Power pin disconnected
3	JP301	Regulated 5 V	Protected switched VBAT	Power pin disconnected

VBAT follows the input voltage, less protection and wiring losses. It is not regulated 12 V. A sensor requiring 10 V minimum can stop operating when the FRC battery sags, even though the board itself continues running. The 5 V rail has a 300 mA design target shared across all ports selecting it; the VBAT rail has a separate 300 mA shared target. These are total rail budgets, not 300 mA per port. Account for sensor inrush and bias currents.

Software switches the two shared rails on/off and reports their voltage and fault state. It does **not** change a port's voltage selection and cannot read the shunt position. Rail mask: 0=off, 1=5 V, 2=VBAT, 3=both. An erased/unconfigured board starts with both rails off. A stored nonzero rail setting can restore power automatically after ordinary startup.

Do not connect an external power source to a port power pin. For externally powered sensors, remove that port's power shunt, connect common ground and signal, and choose an appropriate bias. The battery switch and reverse-polarity MOSFET do not provide general reverse-current isolation.

Select input threshold and bias

Port	Threshold jumper	Bias header
1	JP102	JP103
2	JP202	JP203
3	JP302	JP303

Threshold jumper fitted selects logic mode: target LOW ≤0.8 V and HIGH ≥2.4 V. Jumper removed selects industrial mode: target LOW ≤4 V and HIGH ≥7 V. Voltages between the LOW and HIGH limits have hysteretic/unspecified state. Positive signal range is 0–30 V. This is digital detection, not an analog measurement input; 1.8 V logic is not guaranteed.

Fit at most one bias shunt, horizontally across one row of the 2×3 header:

Bias row	Bridge	Use
Top	1–2	Pull up through 4.7 kΩ to the port's selected supply; usually industrial NPN/open collector
Middle	3–4	Pull up through 4.7 kΩ to 3.3 V; suitable for low-leakage logic open drain
Bottom	5–6	Pull down through 4.7 kΩ; usually PNP sourcing outputs
None	Open	Push-pull output or external bias

For a 10–30 V NPN photoelectric sensor powered by VBAT: select VBAT power, industrial threshold, and top-row bias. For PNP: select VBAT, industrial threshold, and bottom-row bias. For a 5 V push-pull sensor: select 5 V power, logic threshold, and no bias. Check the actual sensor's output type, minimum supply and off-state leakage. Use the software inversion mask if the desired logical meaning is opposite to the electrical level.

Input state is sampled at 1 ms intervals. Pulses shorter than a sampling interval can be missed, and debounce adds delay. The counters follow the firmware sampling/filtering path; they are not hardware edge-capture counters. Choose sensor timing and filtering accordingly.

CAN and service connection

J2/J3 are electrically parallel for daisy chaining. Use twisted CAN H/L wiring and a short local stub. JP2 adds 120 Ω termination; fit it only at a bus endpoint. With power off, a correctly terminated whole bus usually measures about 60 Ω between H and L.

J8 uses the **standard Tag-Connect TC2050-IDC-NL mechanical footprint with a custom electrical pinout**. A generic 1:1 TC2050 cable needs an adapter wired to this table. It is not the standard ARM Cortex 10-pin mapping.

J8/cable pin	Board signal	Connect to
1	NRST	Probe reset, optional

J8/cable pin	Board signal	Connect to
2	UART_RX	3.3 V UART adapter TX
3	UART_TX	3.3 V UART adapter RX
4	SWCLK	SWD probe clock
5	SWDIO	SWD probe data
6	+3V3	Probe VTref **sense input only**
7	Ground	Probe ground
8	Ground	Service ground
9	BOOT0	Leave open normally; use only for deliberate bootloader service
10	Ground	Service ground

Power the board at J1 during programming. Leave programmer/UART power outputs disconnected. Align the cable using its three guide pins; the contacts have 1.27 mm pitch. UART is 115200 baud, 8N1, 3.3 V logic. Secure the cable with a suitable TC2050 retaining clip or fixture; a bare pogo pattern is not a permanent connector.

6 Firmware and programming

Source: firmware/README.md

Scope: Rev B-2. Revision overview · All revisions.

Bare-metal STM32F042K6 firmware for three inputs. The fourth protocol channel is reserved and always zero. Two final 1 KiB flash pages hold redundant CRC-protected configuration. The image uses 7,148 bytes of flash and 1,832 bytes of RAM including reserved stack; these are linker allocations, not measured stack depth.

From `rev_b2/`:

```
source tools/env.sh
make -C firmware
bash scripts/test_firmware.sh
```

The build produces `firmware/build/sensor_can.elf`, `.hex`, and `.bin`. Host tests simulate peripheral boundaries; they do not qualify real STM32 registers or wiring.

Power J1 from a current-limited bench supply. Use the custom TC2050 adapter in WIRING.md: probe VTref sense→J8.6, SWDIO→5, SWCLK→4, ground→7, optional reset→1. Leave probe power disconnected. The mechanical footprint is standard TC2050-IDC-NL; the electrical mapping is custom.

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -c 'adapter speed 1000' \
  -c 'program firmware/build/sensor_can.elf verify reset exit'
```

This has not been exercised on physical Rev B/B-2 hardware. Verify target identity, voltage and flash readback. Normal application updates preserve configuration; a full-chip erase removes it.

For UART, connect a 3.3 V adapter's TX→J8.2, RX→J8.3, ground→J8.8 or 10. Leave adapter power disconnected. UART is 115200 baud, 8N1.

```
python host/sensor_can.py info --port /dev/ttyUSB0
python host/sensor_can.py configure --port /dev/ttyUSB0 \
  --device 17 --invert 1 --debounce 2 --rails 0
```

Device IDs are 0–63 and must be unique for this Team Use protocol. Rev B/B-2 inversion masks are 0–7; bit 0 is sensor 1. Debounce is 1–100 ms. Rail masks: 0=off, 1=5 V, 2=VBAT, 3=both. Voltage selection is physical. UART `SET id invert debounce_ms rails` disables rails, writes configuration, and resets. A stored nonzero rail mask enables those rails after ordinary startup; provision rails=0 until wiring is verified. Avoid repeated persistent writes from robot code.

Unconfigured boards keep both rails off and CAN in standby. Valid configuration enables the original 1 Mbit/s, 29-bit Team Use protocol. Inputs are sampled at 1 kHz; counters report debounced logical rising edges. Input LEDs show logical state. This is not an encoder, IO-Link or analog-input device.

Hardware pulldowns establish off defaults. Clock failure, hard fault, PVD low voltage and a foreground stall longer than 20 ms force rails off; the independent watchdog provides reset recovery. Enabled rails below a measured 4.5 V latch off after startup blanking. Battery light-load FAULT is distinguished using output voltage. Broadcast Disable turns both rails off and clears the requested rail state. Recovery needs an explicit rail request and fault clear after the off-time requirement. Do not continuously retry faults.

CAN initialization failure can leave deliberately configured sensor power requested while communication is unavailable. Robot code must check fresh status, diagnostics and the sensor's supply requirements. Reading data never enables power. See CAN_PROTOCOL.md for packets and recovery policy.

7 CAN protocol

Source: docs/CAN_PROTOCOL.md

Scope: Rev B-2. Revision overview · All revisions.

Rev B-2 has three physical channels. Only inversion/state bits 0–2 are used. The fourth state bit and fourth counter slot stay zero for protocol compatibility. Firmware identity is 0.2.0, hardware 2.

FRC Sensor CAN protocol version 1

The interface uses 1 Mbit/s Classical CAN with 29-bit extended data frames. It is an original Team Use protocol, with device type 10 (Miscellaneous) and manufacturer 8 (Team Use), following the WPILib addressing layout. It does not implement CTRE Phoenix or impersonate CTRE hardware. Coordinate IDs with any other team devices using this same type/manufacturer pair.

```
identifier = (10 << 24) | (8 << 16) | (api << 6) | device_id
```

`device_id` is 0–63, `api` is 0–1023. For example, device 17 status is `0x0A084011`. All team packets below have DLC 8. Multibyte integers are unsigned little-endian. Reserved bytes/bits must be zero. Invalid frame type, length, version, command magic, ranges or reserved bits are rejected. Receive filters reserve a separate FIFO for the standard FRC Broadcast Disable message.

API	Direction	Period	Bytes 0 through 7
0x100 status	Board → robot	10 ms	protocol=1, sequence, logical inputs, raw inputs, enabled rails, latched faults, health flags, reset flags
0x101 counts 1/2	Board → robot	100 ms	channel 1 uint32, channel 2 uint32
0x102 counts 3/4	Board → robot	100 ms	channel 3 uint32, reserved zero uint32
0x110 diagnostics	Board → robot	100 ms	switched battery output mV uint16, sensor 5 V mV uint16, TX queue drops uint16, TEC uint8, REC uint8
0x111 identity	Board → robot	1000 ms	protocol=1, firmware major=0, minor=2, patch=0, hardware=2, device ID, debounce ms, inversion mask
0x200 runtime configuration	Robot → board	On explicit change	0x53, 0x43, 1, inversion mask, debounce low byte, debounce high byte, requested rails, 0
0x201 clear fault	Robot → board	On explicit request	0x53, 0x43, 1, clear mask, 0, 0, 0, 0

Input bits 0–2 mean channels 1–3; bit 3 is reserved zero. Raw inputs are comparator levels before inversion and debounce. Logical inputs are the stable, debounced states after XOR with the configured inversion mask. Rising-edge counters count changes to logical TRUE; they start at zero on boot and wrap modulo 2^{32}. Counter-pair frames are separate snapshots, not an atomic snapshot of all channels. Runtime configuration reinitializes debounce state while preserving counts. Input LEDs show logical state.

Rail bit 0 means regulated 5 V; bit 1 means battery. An enabled bit reports the MCU's commanded switch-enable state, **not measured power-good**. A hardware switch can have shut down before software updates the status. Latched faults require manual recovery. The battery diagnostic measures SENSOR_VBAT after the battery switch. The 5 V diagnostic measures the shared switched 5 V rail. Both are measured on the PCB; cable/connector loss and a disconnected sensor remain outside that measurement. Firmware samples

both every 1 ms and latches an enabled rail off below a measured 4.5 V after its 5 ms startup blanking interval. ADC failure latches both rails off. The battery switch also asserts its digital FAULT for healthy loads below 10–20 mA, so battery fault handling uses output voltage while the switch retains autonomous current-limit/latch protection.

Health flags: bit 0 configured, bit 1 clock healthy, bit 2 ADC healthy. Reset flags: bit 0 power-on, bit 1 reset pin, bit 2 independent watchdog, bit 3 software reset; several bits can be set. Diagnostic ADC readings are approximate monitoring values, not calibrated instrumentation. Ignore voltage readings if ADC health is false. TX queue drops count local mailbox submission failures, saturating at 65535; they do not count every unacknowledged or corrupted transmission. TEC and REC are controller error counters.

Debounce is 1–100 ms with 1 kHz sampling. The default service example uses 2 ms. Analog filtering and sample phase add latency; seven typical-model SPICE cases observed roughly 2–3.1 ms transition latency with this setting. No pulse-width, quadrature, encoder speed or minimum-pulse guarantee is established. The status sequence increments modulo 256 and is a freshness aid, not a transaction identifier.

Runtime configuration accepts inversion 0–7 and rails 0–3. It is volatile and does not clear a latched rail. To recover a fault: remove the cause, leave the rail off for at least 100 ms, explicitly request the desired rail state, then explicitly clear its fault bit. Clear masks are 1–3. Do not retry automatically in robotPeriodic. UART SET is the only normal persistent configuration operation and resets the board after a verified flash write.

Timing and short pulses

Status is a periodic sample, so a brief debounced pulse can occur between 10 ms transmissions. Rising-edge counters retain such events until reset but arrive every 100 ms and have no event timestamps. Local debounce latency also excludes CAN and robot-loop delays. See the sampling and timing notes before choosing a sensor geometry or event-handling strategy.

Broadcast Disable and startup

Extended data identifier **0x00000000**, with any DLC 0–8, is FRC Broadcast Disable. Firmware immediately commands both rails off, sets both latches and clears requested rails. Communication remains available. Recovery requires an explicit rail request and fault clear after the off-time requirement. Ordinary disabled-state heartbeat does not switch sensor power in this sensor-only design. This device has no actuator outputs.

With no valid flash configuration, CAN remains in standby and sensor rails remain off. Assign a unique ID through UART before connecting several unconfigured boards to a robot. A watchdog reset keeps both sensor rails latched off even if stored configuration requests them. Clock or low-voltage faults force safe shutdown. Automatic CAN bus-off recovery is enabled, but sensor data must still be treated as unavailable when stale.

Robot-side rules

Use a 100 ms status freshness limit; this is implemented in the supplied Java and C++ wrappers. Counter/diagnostic freshness is 350 ms and identity freshness is 1500 ms. Missing or invalid packets return no sample. Do not substitute a stale FALSE for a sensor failure. Verify configured, clock and ADC flags, the relevant rail enable/fault bits, and adequate sensor supply before acting. The supplied intake examples fail closed when their assumed 10 V sensor supply margin is absent.

There is no command acknowledgment packet, transaction ID, encryption or authentication. Read status and identity back after configuration. This is intended for a trusted, physically controlled robot CAN bus. Periodic traffic is 131 frames/s; a conservative 160 bits/frame estimate gives about 2.1% bus load per board. Team-controlled command rates should remain low.

Sampling and event timing

Scope: Rev B-2. Revision overview · All revisions.

Inputs are sampled every 1 ms. Firmware applies configurable 1–100 ms debounce, sends status every 10 ms and sends each pair of rising-edge counters every 100 ms. The service example selects 2 ms debounce. There are three active inputs; the fourth protocol slot remains zero.

A short pulse can fall between samples. A pulse detected by one sample can also disappear before the debounce interval completes. Allow at least the configured debounce interval plus a sampling interval, analog/sensor delay and margin; establish the actual minimum pulse width on hardware. A 5 ms pulse is a useful initial bench test with 2 ms debounce, followed by shorter and longer pulses at varied phase. Rev B-2 has no measured pulse-capture guarantee.

A debounced pulse can occur entirely between two status packets. Rising-edge counters retain a captured logical rising event until reset, but their slower reporting adds delay and they carry no timestamp. Select inversion so the event of interest is a rising edge. Establish a counter baseline on startup and after data loss, handle unsigned rollover, and reject implausible jumps after a board reboot. The two counter-pair messages are separate samples.

With 2 ms debounce, local firmware response is roughly 2–3 ms after the sampled electrical transition, excluding analog delay. The next status transmission can add approximately 10 ms and a 20 ms robot loop can add almost another 20 ms. CAN arbitration and transport delay add to this scheduling budget; it is not a measured or guaranteed end-to-end bound.

Check the shortest beam interruption at the highest expected game-piece speed. Test the complete sensor, wiring, board, CAN and robot loop. These filtered digital inputs and telemetry rates are not qualified for PWM, quadrature or high-speed encoder acquisition. The Rev A comparator pulse study is not claimed as Rev B-2 validation.

8 Robot integration

Source: examples/README.md

Scope: Rev B-2. Revision overview · All revisions.

Robot integration — WPILib 2026.2.1

The Java and C++ interfaces read this board's original Team Use CAN protocol. They compile against WPILib 2026.2.1 APIs. Host tests replace the transport boundary; an actual roboRIO integration test remains required.

For Java, copy `java/src/main/java/frc/robot/sensors/SensorCan.java` and `SensorCanCodec.java` into the matching package in a Java 17 robot project. Construct one `SensorCan` per physical board as a subsystem member, using the ID provisioned through UART. `status()`, `diagnostic()`, `counts12()`, `counts34()` and `identity()` return `Optional<Sample<...>>`. Missing, malformed or stale data returns empty. Close the object when disposing the subsystem.

For C++, copy the two headers from `cpp/include` into the robot project and construct `sensorcan::SensorCan` as a subsystem member. `ReadStatus()`, `ReadDiagnostic()`, `ReadCounts12()`, `ReadCounts34()` and `ReadIdentity()` return `std::optional<Sample<...>>`. The class uses the project's normal WPILib CAN library; no additional runtime library is required.

The intake examples in both languages assume channel 1 is logically TRUE when a game piece is present, with a battery-powered sensor requiring at least 10 V. They permit intake only when fresh status and diagnostics exist, configured/clock/ADC flags are healthy, battery power is enabled without a fault, switched battery output reports at least 11 V, and the piece is absent. The 11 V check is an example allowance for switch/wiring drop and diagnostic error; measure your actual sensor and adjust its policy deliberately. Neither example is a complete motor-control subsystem.

Reading data never enables sensor power or clears faults. `configureRuntime` / `ConfigureRuntime` sends one volatile configuration request; `clearFaults` / `ClearFaults` sends one explicit recovery attempt. Do not call either continuously from periodic code. Verify the resulting status/identity, and expose failures to the driver station. A constructor or successful packet write does not prove the board is present.

Status expires after 100 ms; counter/diagnostic packets after 350 ms; identity after 1500 ms. Counters are unsigned 32-bit totals and wrap. Java represents them as `long`; C++ uses `uint32_t`. Counter pairs arrive separately. Do not use these 1 kHz debounced inputs as high-speed encoders or rely on them as a safety interlock.

The Python service tool can also passively monitor a Linux SocketCAN interface already configured for 1 Mbit/s:

```
python host/sensor_can.py monitor --channel can0 --device 17 --seconds 30
```

Configure the CAN adapter using its own supported procedure. The monitor sends no commands and reports unavailable status after 100 ms without a valid frame. UART provisioning is described in firmware/README.md.

Run `bash scripts/test_drivers.sh` from the project root to compile and exercise Java, C++ and Python integrations using the installed toolchain. No LabVIEW VI is supplied; a LabVIEW implementation would need to implement the documented packets, freshness checks and fault policy, then be tested on the robot. This project does not claim complete ecosystem certification.

9 Qualification worksheet

Source: docs/QUALIFICATION.md

Scope: Rev B-2. Revision overview · All revisions.

Status: **not physically built or tested**. Record board serial number, PCB lot, firmware hash, instruments, ambient temperature and operator for every run. Start with current-limited laboratory supplies and controlled loads. Exported design checks establish file consistency; they do not establish a production rating.

Test	Method and acceptance target	Measured result
Assembly	Inspect polarity, MCU bridges, exposed-pad wetting, fuse, terminal double legs, solder-mask coverage and all seven NPTH holes. Confirm no shorts before applying power.	Pending
No-load supplies	Sweep J1 through 5, 6.3, 12, 14.4, 20 and 22 V. Record input current, intermediate rail, internal 5 V and 3.3 V; no oscillation or abnormal heating.	Pending
Full shared-rail loads	Apply 300 mA to 5 V and 300 mA to VBAT simultaneously. Repeat each sensor port carrying the full allowed shared-rail load. Measure voltage at the external terminal, including wire and shunt loss.	Pending
Low-input startup	Cold-start at 5 V at all load combinations and repeat with realistic source/wire resistance. Inspect both switching nodes and ripple.	Pending
Voltage transients	Sweep and interrupt the input; capture MOSFET, TVS and converter voltages with appropriate probes. Confirm absolute maxima and recovery. The TVS rating does not establish load-dump immunity.	Pending
Short/inrush	Test each shared output with controlled shorts and representative sensor capacitance. Verify hardware limit/latch, firmware fault report and explicit clear. Evaluate the 40 µF external VBAT capacitance target at 22 V.	Pending
Thermal	Repeat full-load tests over the intended enclosure/ambient range. Record regulator, rectifier, inductor, switch, fuse and connector temperatures against datasheet limits with margin.	Pending
Input thresholds	For every channel and setting, ramp 0–30 V in both directions and record switching/hysteresis. Verify logic LOW at 0.8 V/HIGH at 2.4 V and industrial LOW at 4 V/HIGH at 7 V over supply/temperature.	Pending
Sensor bias	Exercise NPN, PNP, push-pull and open-contact examples with every valid bias setting. Measure leakage/disconnection behavior. Confirm external outputs tolerate the chosen pull-up voltage.	Pending

Test	Method and acceptance target	Measured result
Backfeed	With board off and rails disabled, apply representative externally powered sensor signals within the specified range. Measure logic and rail backfeed; no unintended powering. Never connect two power sources to a selected sensor rail.	Pending
Firmware failures	Force foreground stall, watchdog reset, PVD/brownout, crystal/CSS failure and a latched output fault. Check documented safe-off behavior and recovery.	Pending
CAN	Test on a correctly terminated 1 Mbit/s robot bus with realistic traffic. Exercise counters, filtering, loss/recovery, duplicate-ID handling and Broadcast Disable. Verify custom Team Use identifiers on a real roboRIO.	Pending
Programming	Repeatedly mate the TC2050 cable, verify orientation, target sense and custom adapter continuity, flash/read back, and inspect pogo-pad wear.	Pending
Mechanical	Confirm terminal wire retention, actuator access, mounting/enclosure clearance, shunt retention and vibration behavior. Review fabrication-house CAM before volume ordering.	Pending

Record corrections and rerun affected checks after any failure. The fixed 5 V converter's tolerance and board drops mean the external output is nominal 5 V; applications requiring a strict minimum need measured limits. VBAT follows the input and is never regulated 12 V. Current limits are shared across ports selecting the same rail.

10 Validation evidence

Source: docs/VALIDATION.md

Scope: Rev B-2. Revision overview · All revisions.

These results apply to the supplied Rev B-2 native board and its exported files. They are computational evidence for an unbuilt prototype. Physical soldering, temperature, surge behavior, real sensor/CAN traffic and programming-cable mating remain untested.

Check	Result	Evidence
Native KiCad 10.0.6 ERC / DRC	No reported violations or unconnected items under the saved project rules	`reports/erc/final.json`, `reports/drc/final.json`
Model / schematic / board parity	154 modeled components, 403 model pins, 87 named functional nets; no mismatches	`reports/design_parity.json`
Fabrication and assembly parity	Nine graphic layers, 307 plated and seven nonplated drills, 144 purchased placements and 144 local model assignments	`reports/manufacturing/verification.json`
Independent copper reconstruction	No shorts, opens or unidentified copper islands	`reports/manufacturing/gerber_connectivity.json`
Independent manufacturing geometry	Copper clearance, edge/mount spacing, annular rings, hole web, mask/paste checks passed	`reports/manufacturing/gerber_geometry.json`
Programming contact geometry	Ten contacts, no paste, minimum foreign trace/via clearance 0.751 mm versus 0.508 mm requirement	`reports/programming_geometry.json`
Electrical component corners	8,192 evaluations in each input mode, power/protection assertions passed	`reports/electrical/calculations.json`
Ground-loaded input thresholds	All channels and both modes pass 108 supply/load allocations with correlated ground potentials	`reports/electrical/loaded_input_thresholds.json`
Ground sheet estimate	0.10 / 0.075 mm meshes, analytic calibration and reciprocity/residual checks	`reports/electrical/ground_return_model.json`
Boost startup model	Approximately 4.99 V at 420 mA and 462 mA modeled loads	`reports/boost_simulation.json`
Firmware	Builds for STM32F042; 2,190,989 core assertions plus 19 application fault/startup scenarios pass with sanitizers	`reports/firmware-build.log`, `reports/firmware-tests.log`
Host / robot drivers	232,928 Java assertions, 75,828 C++ assertions, five Python tests; WPILib 2026.2.1 compile checks	`reports/driver-tests.log`

Electrical results

Mode / channel	Rising threshold range (V)	Falling threshold range (V)
Logic / 1	1.569–2.261	0.937–1.447
Logic / 2	1.496–2.240	0.863–1.425
Logic / 3	1.529–2.240	0.896–1.425
Industrial / 1	5.349–6.784	4.715–5.966
Industrial / 2	5.336–6.779	4.702–5.961
Industrial / 3	5.215–6.778	4.581–5.959

The fine-grid estimate gives a worst sensor-ground-to-input-ground resistance of 20.28 mΩ at 85 °C copper temperature. The maximum route change between the last two meshes is 2.43%. Loaded-threshold analysis inflates the modeled ground potentials by 15%, tests both rail load extremes at all port allocations and uses 5, 12 and 22 V inputs. It includes a conservative 150 mA logic allocation and 80% conversion-efficiency assumptions. Switching-frequency ground bounce, external wiring drop and hardware tolerances outside the stated allowances are not represented.

The nominal 5 V output budget includes the converter minimum, actual routed positive-trace resistance, 135 mΩ switch resistance and 100 mΩ for contact/shunt/ground allowance. Its estimated minimum is approximately 4.75 V at a port carrying the whole 300 mA shared load. The input calculation uses 110 mΩ for trace/connector/return allowance in addition to the fuse and MOSFET. These are budgets requiring measurement, not production limits.

The boost runs use TI’s typical transient model and a finite-resistance upstream source. The 3.80 V / 462 mA case reports 4.990 V average, about 3.1 mV modeled steady-state ripple and 0.753 A peak inductor current. It does not establish stability of the complete cascaded converters or thermal performance. The original vendor model, syntax adaptation, decks, sampled data and plots are included.

Interpretation and configured checks

Native DRC keeps clearance, connectivity, courtyard overlap, holes, mask and silkscreen checks enabled. Informational/model-category checks disabled in the project are: missing_courtyard, track_not_centered_on_via, tuning_profile_track_geometries, footprint_filters_mismatch, footprint_type_mismatch. Native ERC disables: single_global_label, four_way_junction, simulation_model_issue, footprint_filter. The separate parity and fabrication checks complement these settings. No clearance or unconnected-net violation is waived.

Excellon output rounds coordinates and diameters to 0.001 mm. The independent audit accepts at most half that unit and records the actual rounding error. Gerbonara reports KiCad’s G90 header-placement warning; drill positions, diameters and plating types still match. Polygon arc approximation is 1 µm; reported annular-ring minima are interpreted within that numerical tolerance.

Firmware occupies 7,148 bytes of flash and 1,832 bytes of RAM including the reserved stack allocation. Host tests execute the application with controlled peripheral substitutes; they do not emulate all STM32 electrical behavior. The fourth protocol slot remains reserved and zero on this three-channel board. Actual SWD flashing, roboRIO traffic, fault currents and sustained timing must be checked on hardware.

Historical draft reports and routing experiments in the development workspace may contain failures. The delivery archive includes the final reports identified here. Complete `docs/QUALIFICATION.md` before promoting the design from prototype to a qualified product.

B-2 variant and assembly checks

`reports/variant_verification.json` checks the frozen horizontal Rev B signature against B-2. Copper tracks, vias, filled zones, footprint positions and pad geometry are identical. Firmware source/header files and delivered ELF/HEX/BIN hashes are identical. The existing firmware and robot-driver tests are inherited on that basis; these are not physical factory tests.

The schematic, native DRC/ERC, Gerbers, drill/placement parity, ground model and loaded input-threshold checks were regenerated for B-2. Exactly 49 fitted MPNs change (37 resistors and 12 capacitors); their values, tolerances and circuit connections remain fixed, and the capacitors' voltage rating increases. `PART_CHANGES.md` records the specification review.

`reports/factory_placement.json` verifies the 144-reference BOM/CPL mapping, coordinates, and six horizontal connector drill patterns. There are 128 SMT parts/338 solder joints and 16 THT parts/65 solder joints. The CPL uses a lower-left origin and explicit through-hole body-center offsets. Factory library rotation, rail/tab design, sourcing and build-process approval have not been performed by a supplier.

11 Capacitor review

Source: docs/PASSIVE_REVIEW.md

Scope: Rev B-2. Revision overview · All revisions.

This design has 34 capacitors. They are not all interchangeable filtering capacitors, and the present count is not proven to be the absolute minimum. This audit retains them until a reduction has measurement evidence.

References	Job	Decision
C1 C2 C3	Battery/input filtering	Keep
C4 C5 C13	Buck output reservoir	Keep; C13 is a bench optimization candidate
C6	Buck bootstrap	Keep
C7 C8	3.3V regulator input/output	Keep
C9 C10	Battery sensor switch bypass/output	Keep
C11 C12	5V sensor switch bypass/output	Keep
C14	Hardware fault timing	Keep
C15 C16 C17	Boost output reservoir/bypass	Keep
C20 C21 C22 C23	Microcontroller supply bypass	Keep
C24	Microcontroller local bulk	Keep; bench optimization candidate
C25 C26	Crystal load	Keep
C27	Reset filtering	Keep
C28 C29	Supply measurement filtering	Keep
C40 C41	CAN supply bypass	Keep
C42	CAN local bulk	Keep; bench optimization candidate
C50	Comparator supply bypass	Keep
C101 C201 C301	Sensor input filtering	Keep

Several 100 nF capacitors share a supply net because they sit beside different chips or supply pins. Copper traces have inductance: one distant large capacitor cannot supply every fast local current pulse. Other capacitors are part of the converter, crystal, fault timer or input filter rather than rail bypass.

C13, C24 and C42 are candidates for a measured reduction experiment. Test supply extremes, both sensor rails loaded, startup, load steps, ripple, CAN traffic, reset behavior and temperature before approving their removal. No capacitor has been removed on the basis of appearance or a DC continuity check.

Most small resistors set input thresholds/hysteresis, sensor pull-up or pull-down current, regulator output voltage, fault current limits, safe enable states, ADC scaling or LED current. Many small black parts in the render are resistors, not capacitors. A factory-assembly variant can use lower-cost stocked equivalents without deleting these functions.

The larger simplification opportunities are the input-configuration network and the two-stage 5 V supply architecture. These require a circuit redesign and validation; changing to smaller package sizes alone does not remove their functions. See capacitor_audit.csv in the BOM folder for every capacitor and its reason.

12 Sources and notices

Source: docs/SOURCES_AND_LICENSES.md

Scope: Rev B-2. Revision overview · All revisions.

Sources were consulted on 2026-09-07/08. Exact purchase SKUs, product-page links, stock observations and procurement prices are stored in `bom/catalog_china.json` and `BOM_grouped.csv`. They are dated distributor observations, sometimes from indexed listings, not reserved inventory or proof of worldwide delivery. The B-2 BOM uses LCSC and DigiKey sources. Its five incorporated substitutions are reviewed in PART_CHANGES.md; further substitutions require engineering review.

Primary technical references

Subject	Primary source	Used for
Functional reference	CTRE CANdi, user guide	Sensor-to-CAN role and regulated/battery sensor outputs; no inherited ratings/protocol
Robot electrical rules	FIRST 2026 manual	Protected power branch, wiring, isolation and CAN constraints; recheck season rules
CAN addressing	WPILib CAN device addressing	Team Use manufacturer, device type, Broadcast Disable
Robot CAN API	WPILib 2026.2.1 source, Java CAN API	Actual Java/C++ transport interfaces
MCU	STM32F042x4/x6 datasheet, ES0243 errata	LQFP32 pin map, voltage/clock/ADC limits, calibration delay
CAN PHY	TCAN3413 datasheet	SOIC8 pin map, supply, standby and bus protection
Buck regulator	LMR51420 datasheet	X=500 kHz PFM variant, L/C, reference, layout and current limits
Logic regulator	AP2112 datasheet	SOT25 pin map and output-capacitor requirements
Battery switch	TPS1H200A-Q1 Rev. E	DGN pin map/land pattern, limit, latch delay, diagnostics
5 V switch	TPS2553 datasheet	DBVR-1 latch variant, current-limit equations, fault timing
Comparator	LM339LV datasheet	3.3 V operation, failsafe input range, quad pin map
Boost	TPS61322 datasheet, TI transient model	Fixed 5 V converter, limits, startup model
Boost inductor	SRP4020TA	4.7 µH ratings and custom land dimensions
Reverse protection	AO4485	MOSFET pinout, gate/drain ratings, resistance and pulse limits
Fuse	Littelfuse 451/453 datasheet	1.5 A package, interrupt rating, resistance and nominal melting integral
Bias diode	BAT46W	100 V reverse rating, forward drop and package
CAN protection	ESD2CAN24-Q1	SOT23 H/L/GND mapping and bus protection
Inductor	SRP7028A	10 µH current ratings and mechanical outline
Crystal	ABM3B	8 MHz, 18 pF load, ESR/C0, drive level and outline
Connectors	Weidmuller 1824420000, 1824430000	Exact 3.5 mm push-in geometry, two solder legs per pole, horizontal /90 entry
Programming contact	Tag-Connect TC2050-IDC-NL	Contact pitch, alignment holes, cable envelope and copper clearance; custom electrical mapping
Capacitor derating	Samsung CL31A226KAHNNNE manufacturer characteristics; original Rev A research file retained in the development workspace	30 µF effective capacitance is an engineering simulation assumption for 44 µF nominal; confirm actual bias/temperature loss
Aluminum capacitor	Panasonic FK series	EEE-FK1H470P dimensions, ripple/impedance and handling cautions
ESP32 comparison	Espressif ESP32-C3 TWAI	CAN controller still requires an external transceiver

Local datasheets and extracted text are research references. Earlier research remains in the parent development workspace; only the current BOM and circuit determine what to purchase. The release manifest

records included source-file hashes. Changed-part PDFs are retained in `sources/`; the research file named `weidmuller_3p_90.pdf` is a rejected different-pole-count variant and is excluded from delivery. Manufacturer documents and models retain their owners' copyright and terms; they are not relicensed as original project material.

Libraries and software

KiCad standard footprints and package models came from the verified KiCad 10.0.6 distribution. Their original names are retained in each part's `source_footprint` field. Project-local copies make the design portable. The KiCad library license is CC BY-SA 4.0 with its design-output exception; the notice and legal text are retained in `licenses`. Original custom footprints cover the selected Weidmuller terminals and TI exposed-pad package. The Abracon crystal footprint adapts the KiCad body/model to the manufacturer's recommended land pattern, with its derivation recorded in the footprint provenance file. Original simplified STEP models retain their provenance; changed Rev B-2 models are listed in `hardware/library/3d/rev_b_models.json`. They are not supplier-certified models.

Firmware includes ST CMSIS device headers/startup and Arm CMSIS headers with their original license files. WPILib headers/jars are used for integration compilation and retain the WPILib notice. InteractiveHtmlBom generates the standalone assembly page; its license is included. TI's original TPS613222A model and copyright header are preserved. The ngspice-compatible copy changes initial-condition syntax, and the test deck supplies the startup parameter documented in the engineering report. See `licenses/` and the respective third-party directories for the complete notices.

No blanket public license has been selected for the user's original hardware, firmware or documentation in this repository. Third-party licenses continue to apply to their own material. The project is versioned in Git; no physical board has been built and no manufacturing order has been placed. CTRE, FIRST, WPILib, DigiKey and component names identify compatibility targets or sources, not endorsement.

B-2 procurement and substitutions

The underlying Rev B circuit and firmware are retained. Same-value substitutions are documented in PART_CHANGES.md, using the UNI-ROYAL manufacturer resistor-family data sheet and Samsung exact capacitor specification. The Samsung sheet is included in `sources/samsung_CL21B104KCFNNNE.pdf`. UNI-ROYAL's PDF was read through the web viewer; direct download returned HTTP 403, so it is linked rather than claimed as an included local PDF.

LCSC prices/stock and JLCPCB fees were checked 2026-09-08 UTC. URLs and dates accompany the BOM and cost model. They are dated observations, not reservations. The package is not affiliated with CTRE, Weidmüller, JLCPCB or LCSC.

Horizontal connector CAD refresh

Official Weidmüller STEP geometry now replaces the simplified /90 terminal envelopes. The original supplier files, transformed-model hashes, public source URLs and alignment method are retained in vendor_connectors.json. Geometry is translated only; display colors are project-assigned. Supplier material retains its original ownership and is not relicensed by this repository. See CONNECTORS.md. Other custom models remain nominal project envelopes.

Connector geometry and sourcing

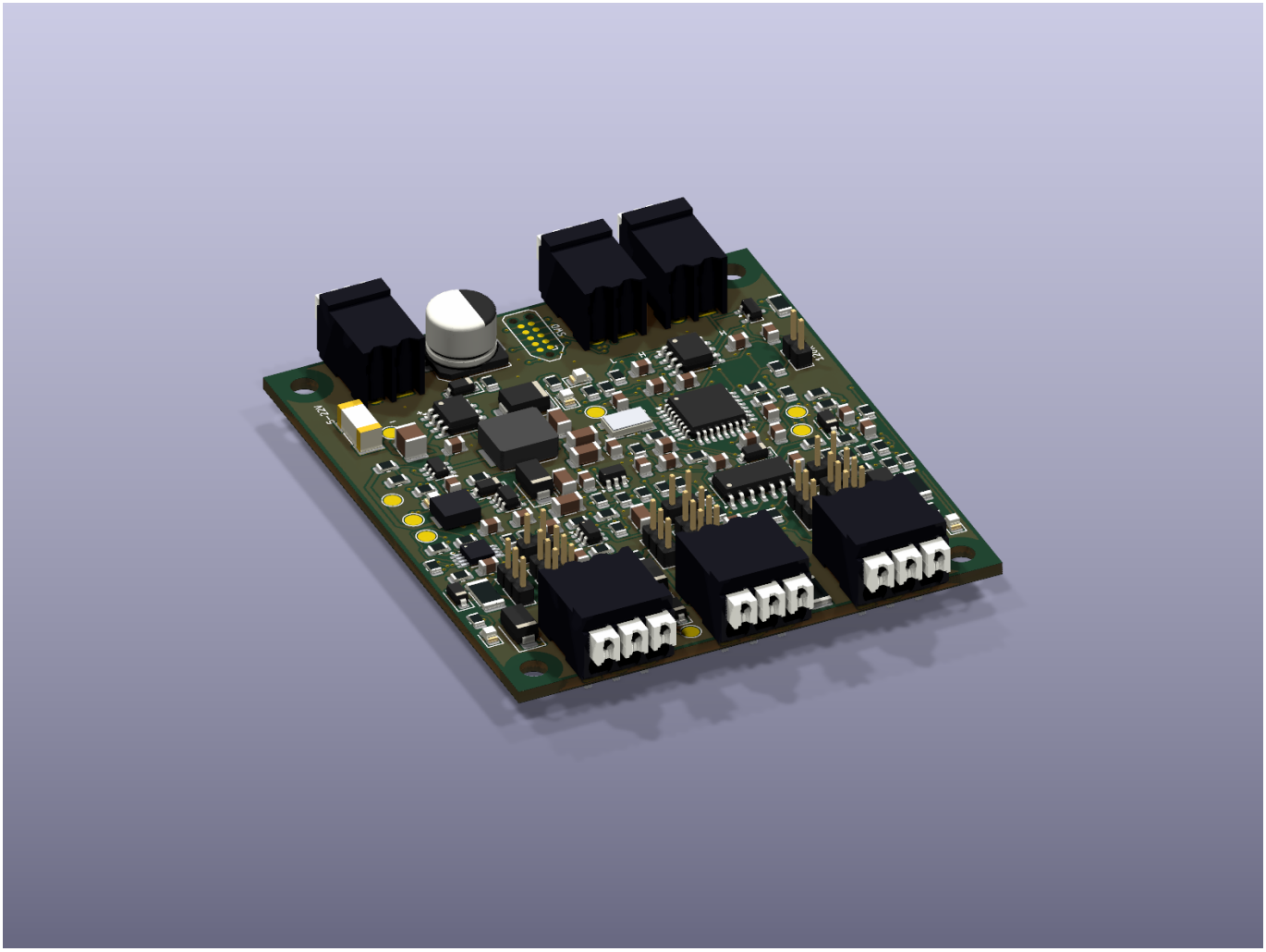
Source: docs/CONNECTORS.md

Scope: Rev B-2. Revision overview · All revisions.

Board references	Manufacturer order code	Full designation	DigiKey code
J1, J2, J3	1824420000	LSF-SMT 3.50/02/90 3.5SN BK TU	281-2020-ND
J4, J5, J6	1824430000	LSF-SMT 3.50/03/90 3.5SN BK TU	281-2021-ND

Both are Weidmüller OMNIMATE LSF-SMT 3.50 push-in terminals with white push buttons and black housings. `/90` identifies horizontal wire entry parallel to the PCB. `/135` is angled and `/180` is vertical; those are not the current parts. The selected terminals have THT/THR solder legs, despite the series name containing “SMT”.

Manufacturer product pages and photographs: 2-pole 1824420000, 3-pole 1824430000. Local drawings: 2-pole datasheet, 3-pole datasheet.



Why the earlier render looked wrong

The earlier project models were simple dimensional boxes with guessed button shapes. The purchasing codes and horizontal selection were correct, but those envelopes did not depict the real housing, buttons or wire openings accurately.

The current board renders and mechanical STEP now use **official manufacturer STEP geometry** downloaded through Weidmüller's public Support Center. Original files are retained unchanged under `hardware/library/3d/vendor/`. A rigid translation places rear pin one at the footprint origin and the seating plane on the PCB. Display colors are assigned by the project to match the black housing, white actuators and metallic pins. The geometry has not been redrawn or rescaled.

The checked STEP pair provides 3.5 mm pole pitch, 7 mm pin-row spacing, two legs per pole, 3.5 mm pin length and 8.5 mm height above the PCB. All ten unique modeled pin centers align with their footprint pads; the pin cross-section fits within the nominal 1.1 mm drill. This is a CAD check, not a measured production-part fit test. The existing native PCB, connector footprints and BOM codes are unchanged.

The vendor CAD's body has a small nominal offset from the generic drawing; its transformed geometry fits inside the existing footprint courtyard and the 14.75 mm reserved depth. Continue reserving **60 × 76.1 mm**, plus wires and actuator access, for the complete board assembly. Supplier CAD is nominal and is not a substitute for checking actual samples and tolerances.

Reproduce and inspect

Run `python scripts/import_connector_models.py` in this revision directory with its Python environment activated. The script verifies the pin pattern, drill fit and vertical envelope before exporting the aligned models. Provenance, hashes and pin coordinates record the exact source files, public URLs and translation.

Regenerate the PCB renders and assembly STEP after importing the models. Manufacturer names and CAD remain third-party material; the project's original envelope models are no longer the geometry used for these horizontal terminals.